4 April '25

# Light- Reflection and Refraction

- **Light** → It is a form of energy which creates a sensation of vision.

Or

It is a form of energy which helps us to see the object clear.

- **Properties of Light**

→ Light is a transverse wave that does not require a medium to propogate.
→ Light travels in a form of ray.
→ Light is a Electromagnetic wave in nature.
→ The speed of light change when there is a change of medium.
→ Light show duel nature that is wave nature and particle nature.
→ Light always travels in a straight path. (Except - defraction)

- **Reflection of Light**

Bouncing back of light in the same medium after striking reflecting surface.

- **Law of Reflection**

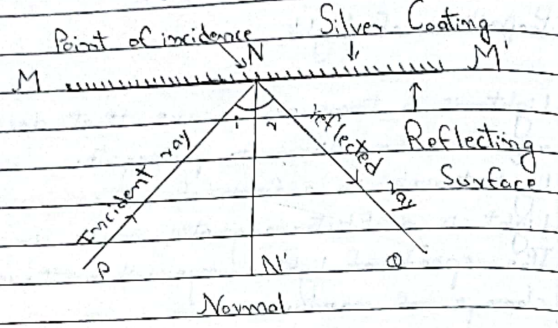
The angle of incidence is always equal to the angle of reflection at the point of incidence. $L_i = L_r$

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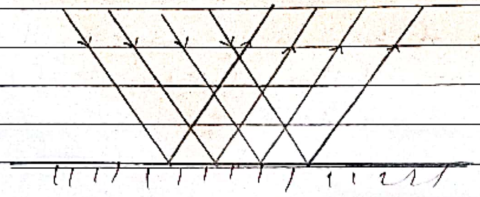
- **Second Law of Reflection**

At the point of incidence, the incidence ray, the reflected ray and the normal all lie in the same place.

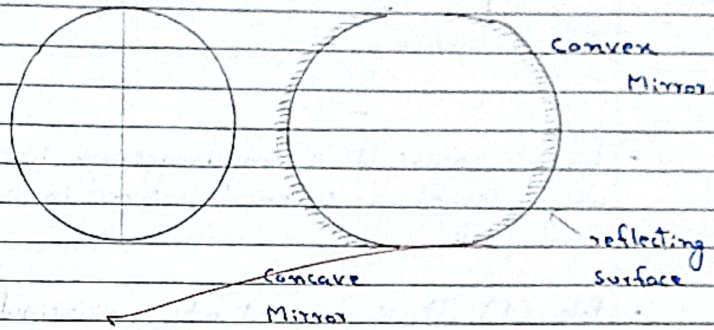
- **Reflection of light**



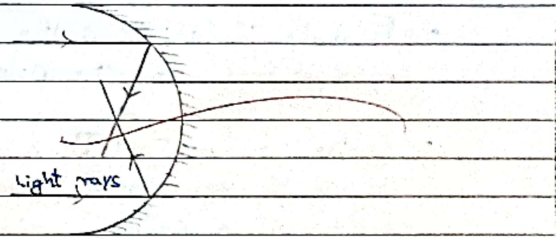

MM' is a plane mirror fixed horizontally with the reflecting surface downwards. A ray of light PN striking with the mirror is called incident ray. This ray striking the mirror at N called the point of incidence. The line NN' draw perpendicular to MM' at point N is called Normal. The ray NQ which bounces back from point N is called the reflected ray.



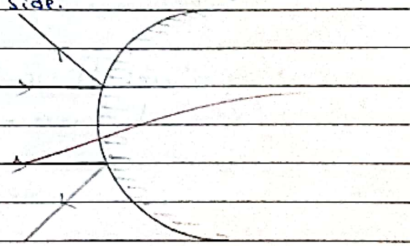
- **Spherical Mirror:** It is a hallow spherical glass which when devided at the centre gives us too different types of surface



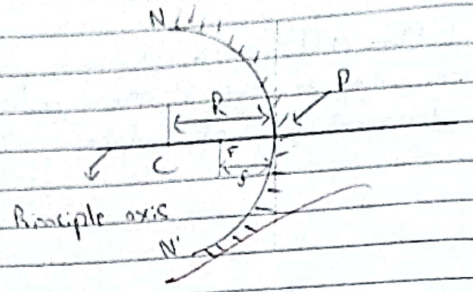

- **Concave Mirror:** A mirror which is polished from the outer side of a hallow sphere, such that the reflecting side is toward its inner side.




- **Convex Mirror:** A mirror which is polished on the inner side of the sphere such that the reflecting surface is toward its bulging side.



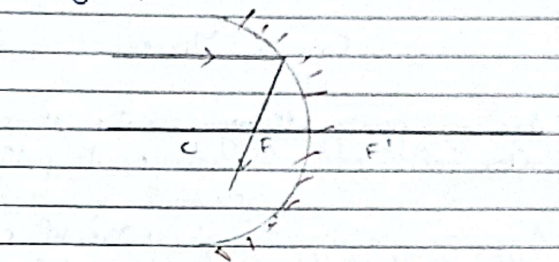
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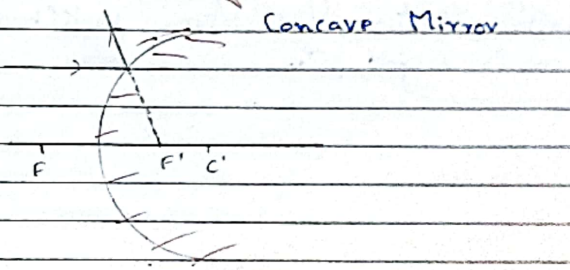

- Principle axis: It is an imagenary line which devides a spherical mirror in two equal parts.
- Pole (P): It is a point where principale axis intersects the mirror.
- Principale focus: It is a point where the reflected ray meet or apear to meet.
- Focal Length (f): It is the distance between principal focus and the pole.
- Centre of Curvature: It is the centre point of a hollow spherical glass of which the mirror is a path.
- Radius of Curvature: It is the distance between centre of curvature and pole of the mirror.
- Aperture: It is the reflecting surface of a spherical mirror.

## Rules for Drawing Ray Daigrams

Rule 1: Any ray travelling parallel to principale axis, after reflection will pass or apear to pass through principle focus

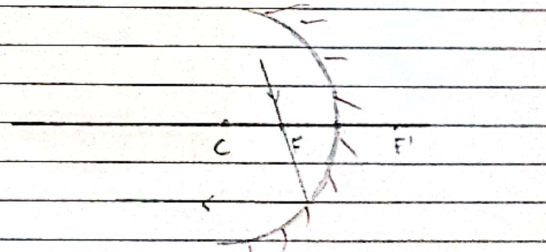



Concave Mirror



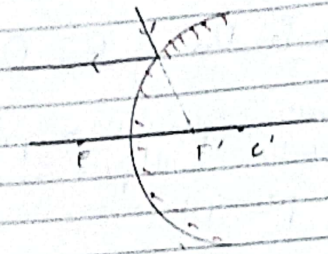

Convex Mirror

Rule 2 Any ray passing through focus, after reflection, will travel parallel to principal axis




Concave Mirror

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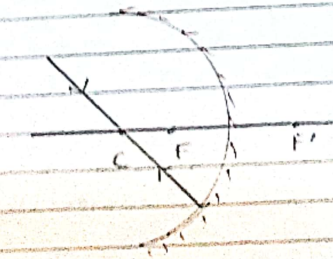



Convex Mirror

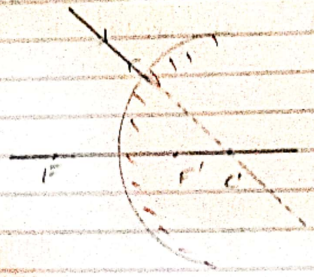
Rule 3 Any ray passing through centre of curvature, after reflection will retrace its path back

or

Any ray passing through centre of curvature after reflection, will return back (bounce back) on its original path.




Concave Mirror




Convex Mirror

## Image formation by Mirror Eg

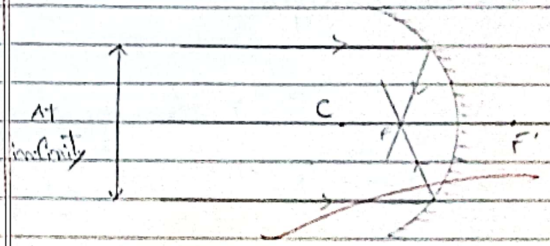
- Concave Mirror:

1. object - At infinity (∞)

Image - At focus

Nature - Real & inverted

Size - Highly Diminished, Point sized

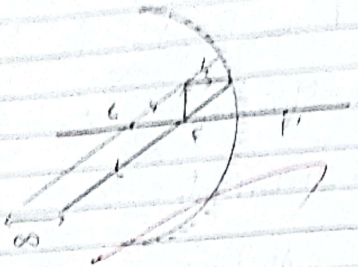



2. object - At focus

Image - At Infinity

Nature - Real & Inverted

Size - Highly Enlarged, Highly magnified

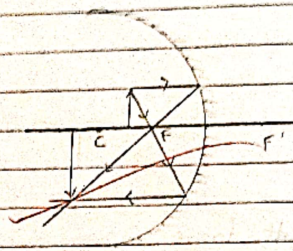



3 Between
object - Beyond F and C

Image - Beyond C

Nature - Real & Inverted

Size - Magnified, enlarged

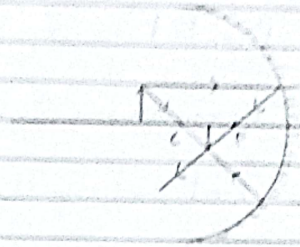


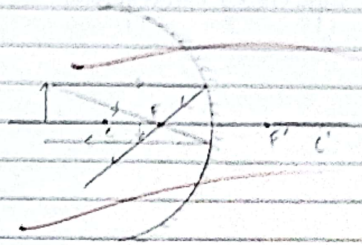

4 Object - Beyond C

Image - Between F and C

Nature - Real & inverted

Size - Diminished



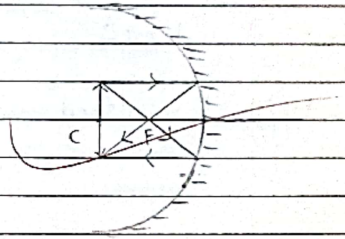




5 Object - At C

Image - At C

Nature - Real & Inverted

Size - Same Size



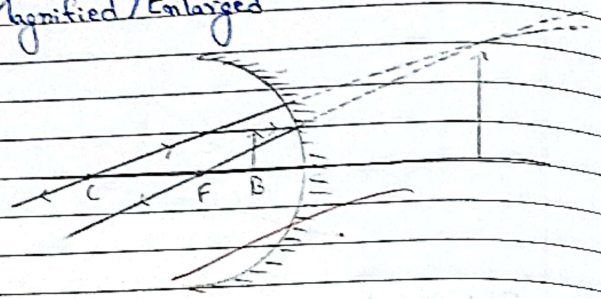

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6 Object - Between P and F

Image - Behind the mirror

Nature - Virtual & erect

Size - Magnified / Enlarged



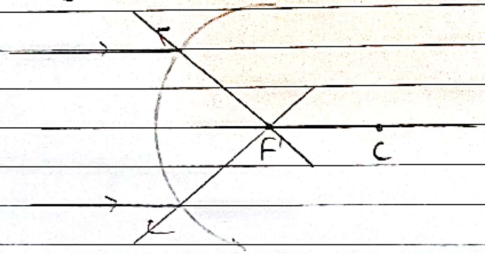

- Convex Mirror:

1 Object - At infinity

Nature - At focus

Image - Virtual & erect

Size - Highly Diminished



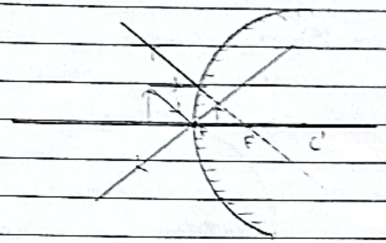

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2 Object - Between Infinity and Pole

Image - Between P and F

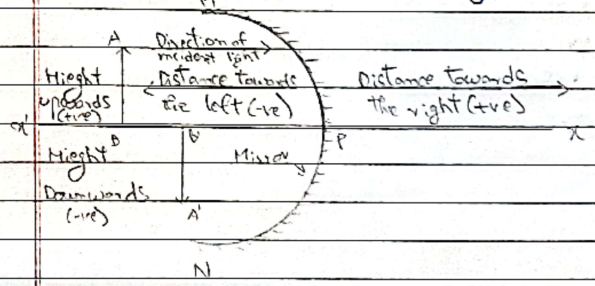
Size - Diminished

Nature - Virtual and erect

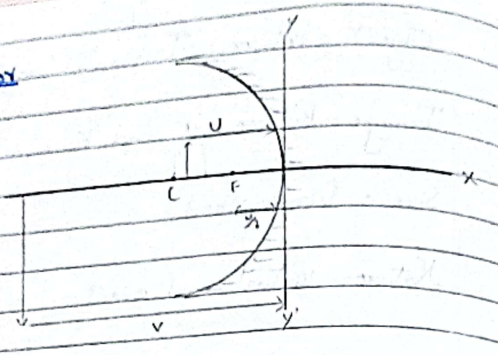



## Sign Convention by Spherical Mirror

f = focal length
u = object distance
v = Image distance
$h_o$ = Height of the object
$h_i$ = Height of the Image

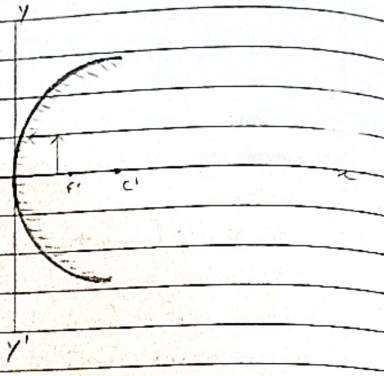


## Concave Mirror




u = -ve
v = -ve
f = -ve
$h_i$ = -ve
$h_o$ = +ve

## Exception of Concave Mirror




u = -ve
v = +ve
f = -ve
$h_i$ = +ve
$h_o$ = +ve

## Convex Mirror:

u = -ve
v = +ve
f = +ve
$h_i$ = +ve
$h_o$ = +ve

## Mirror Formula:

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

f → focal length
v → Image Distance
u → object Distance

## Magnification (m):

$$m = -\left(\frac{v}{u}\right)$$

No SI unit because it is a ratio between two similar quantities

$$m = \frac{h_i}{h_o} = \frac{h'}{h}$$

## Questions

Q1 An object is placed at 30 cm in front of a concave mirror of focal length 20 cm. Calculate the image distance, also calculate the magnification produced.

Solu Concave Mirror:

u = -30
f = 20
v = ?
m = ?

Applying Mirror formula:

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$\frac{1}{20} = \frac{1}{v} - \frac{1}{30}$

$\frac{1}{v} = \frac{1}{20} + \frac{1}{30}$

$\frac{1}{v} = \frac{3+2}{60}$

$\frac{1}{v} = \frac{5}{60^{12}}$

$\frac{1}{v} = \frac{1}{12}$

$v = 12$

Magnification $= -\left(\frac{h_i}{h_o}\right)$

$= -\frac{2}{3}$

---

Q2 An object of height 20 cm is placed in front of a concave mirror. After careful observation the height of image is found to be 45cm. Calculate the magnification produced.

Solu $h_i = -45$ cm

$h_o = 20$ cm

$\therefore m = \frac{h_i}{h_o} = \frac{-45^9}{20^4} = \frac{-9}{4} = -2.25$ cm

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**Refraction:**

Deviation of light from it's original path when the medium of light changes.

**Laws of Refraction:**

a The incident ray, the reflected ray and the normal at the point of incidence all lie in the same plane.

b The ratio of sin of incidence to the sin of refraction is a constant.

i.e $\frac{\sin i}{\sin r} = \text{constant}$

**Interface:**

It is an imaginary surface which seperates two different mediums.

**Lens:**

It is a transparent medium made of glass which converges or diverges the light rays.

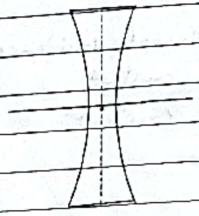
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## Types of lenses:

a Convex lens -

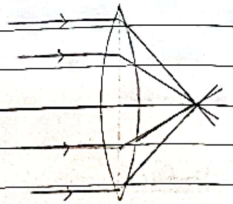


b Concave lens -

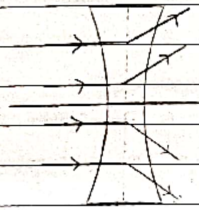


## Properties of Lens:

1) Convex lens -
It is also known as converging lens. The image formed by convex lens is real.

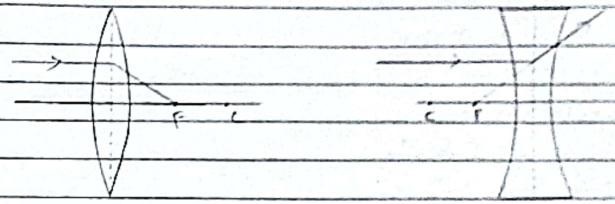


2) Concave lens -
It is also known as diverging lens. Image formed by concave lens is virtual.

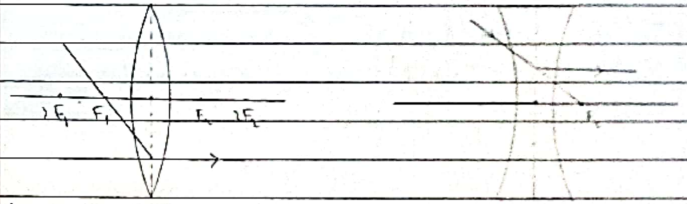


## Rules for drawing a ray diagram -

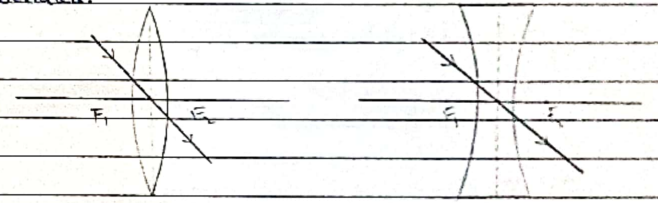
1) any ray travelling parallel to the principle axis, after refraction will pass or appear to pass through focus




2) Any ray passing through focus after refraction will travel parallel to the principale axis.



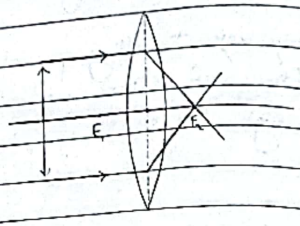

3) Any ray passing through optical centre passes undeviated.



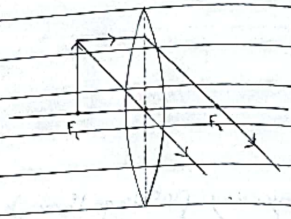

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## Image formation by convex lens –

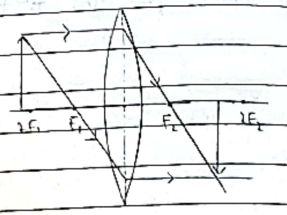
1) Object: At infinity
Image: At focus
Nature: Real & Inverted
Size: Highly diminished / Point size



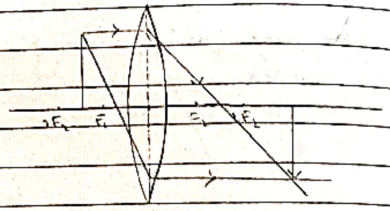

2) object: At focus
Image: At Infinity
Nature: Real & Inverted
Size: Highly Magnified



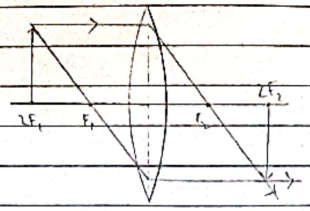

3) Object: Beyond $2F_1$
Image: Between $F_2$ & $2F_2$
Nature: Real & Inverted
Size: Diminished



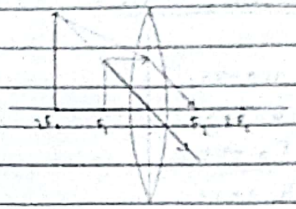

4) Object: Between $F_1$ & $2F_1$
Image: Beyond $2F_2$
Nature: Real & Inverted
Size: Magnified




5) object: At $2F_1$
Image: At $2F_2$
Nature: Real & Inverted
Size: Same sized

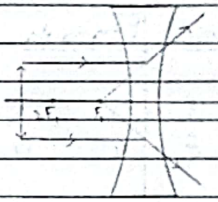



6) Object: Between O and $F_1$
Image: Same side as object (Beyond $2F_1$)
Nature: Virtual & erect
Size: Magnified

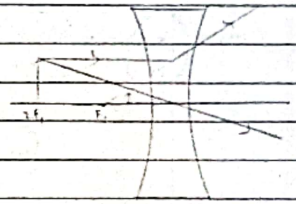



## Image formation by Concave lens –

1) Object: At infinity
Image: At focus ($F_1$)
Nature: Virtual & erect
Size: Highly diminished

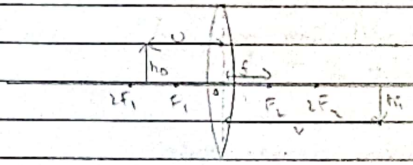



2) Object: Between infinity and optical centre
Image: Between O and $F_1$
Nature: Virtual & erect
Size: Diminished




## Sign Convention of lens:-

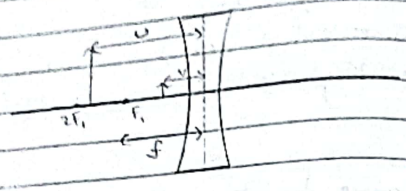
u = -ve
v = +ve
f = +ve
$h_i$ = -ve
$h_o$ = +ve
m = -ve




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## Concave Lens:

u = -ve
v = -ve
f = -ve
$h_i$ = +ve
$h_o$ = +ve
m = +ve




## Lens Formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

## Magnification

$$m = \left(\frac{v}{u}\right) = \frac{h_i}{h_o}$$

## Questions

Q1 The focal length of a convex lens is 20cm. An object of height 5cm is placed at a distance of 30cm from the lense. Calculate:

a Image distance
b Magnification produced
c Height of Image

Solution Given Conven lense

f = 20 cm  $h_i$ = ?
u = 30 cm  v = ?
$h_o$ = 5cm  m = ?

Applying lens formula

$$\rightarrow \frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\rightarrow \frac{1}{v} = \frac{1}{20} - \frac{1}{30}$$

$$\rightarrow \frac{1}{v} =$$

Refractive Index (RI) → Optical density of a medium
It is the optical density of a medium.

→ Absolute RI
u.s $\frac{c}{v}$ = $\frac{3 \times 10^8}{v}$
- Solid (gas)
- Liquid (water)
- Gas (atmosphere)
- Plasma (jelly / glycerine)

→ Relative RI
u.s $\frac{v_1}{v_2}$ → speed of light in medium 1 / speed of light in medium 2

Power of a lens

P.s $\frac{1}{f}$ (in meter)

P.s $\frac{100}{f}$ (in centimeter)

It is the reciprocal of focal length in metre.
SI unit → Dioptre (D)

Power of
a. Concave lense → -ve
b. Convex lense → +ve

(u.s.